Horizontal Pod Autoscaler (HPA) in Kubernetes

Understanding Kubernetes for Orchestration

Introduction

Horizontal Pod Autoscaler (HPA) automatically scales the **number of Pods** in a Kubernetes Deployment based on CPU or memory usage.

This guide will cover:

- **Why HPA is needed?**
- **How HPA works?**
- **Deploying HPA in Kubernetes**
- **Common issues and FAQs**

Why Use HPA?

- **Automatically scales applications** during high traffic.
- **Optimizes resource usage** and reduces costs.
- **Prevents application failures** due to resource shortages.

Step 1: Install Metrics Server

HPA requires the **metrics server** to fetch resource usage data.

```
kubectl apply -f
https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
```

Verify the metrics server:

```
kubectl get pods -n kube-system | grep metrics-server
```

Step 2: Deploy an Application with Resource Requests

HPA needs Pods to have **CPU requests and limits** defined.

Example Deployment:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        resources:
          requests:
            cpu: "100m"
          limits:
```

```
        cpu: "200m"
```

Apply the deployment:

```
kubectl apply -f deployment.yaml
```

Step 3: Enable HPA for the Deployment

Create an HPA resource for the **nginx-deployment**:

```
kubectl autoscale deployment nginx-deployment --cpu-percent=50 --min=2
--max=10
```

Check HPA status:

```
kubectl get hpa
```

Common Issues and Troubleshooting

Problem	Solution
HPA not scaling	Ensure metrics server is installed
`unknown` CPU usage	Check `kubectl get --raw /apis/metrics.k8s.io/v1beta1/nodes`
HPA scales too fast	Adjust `--cpu-percent` threshold

Conclusion

HPA ensures **dynamic scaling** based on resource usage, improving application reliability and cost-efficiency.

👉 Test **HPA by deploying a load generator and monitoring Pod scaling!**